

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY

Spatial informatics

Time: 90 minutes

Total Marks: 45

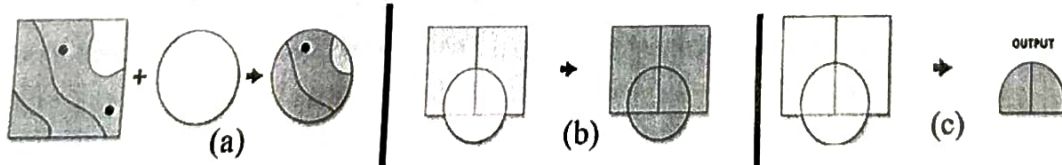
Section-I (11 marks)

1. How do you identify whether an anti-poaching site is located within forest area or outside of it
2. Define Topology as per *Longley*.
3. What is "proximity" in raster data operations.
4. Why should we fix the node errors?
5. What type of generalization technique you will use to represent a dense stream network on a map?
6. What is rubber sheeting?
7. Elucidate the application of "distance-based site selection" under global operations using an example.
8. How are positional accuracy errors caused by ionospheric and tropospheric delays corrected?
9. What are the issues with performing vectorization without prior preprocessing?
10. A map distance of 4 cm represents 2 km on the ground; what is the representative fraction (RF) scale of the map
11. Classify the following operations (a) and (b) (local, focal, zonal, global)

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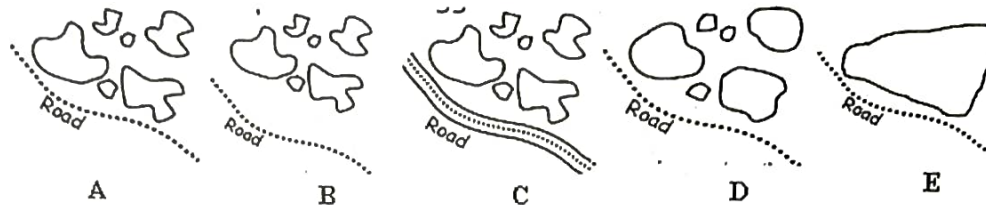
Section II (24 Marks)

12. While relocating his office, Mr. X found an old city map in a desk drawer. Because the map has stretched and distorted over time, list what map properties will change and what properties will remain unchanged?
13. Give one example of each type of topological relationship between: (a) point and area, (b) line and area, and (c) area and area.
14. Identify the following operations and briefly mention when it is useful. Also state which attributes are present in the final output



15. In digital image processing, spatial filtering relies heavily on defining a local neighbourhood around a pixel. Comment on any three common neighbourhood shapes used in spatial operations, providing examples of their typical window structures.
16. Describe any two raster overlay rules with suitable examples.

17. In the following, figure A is the true representation of certain features. Identify and briefly explain the other figures with their generalization methods



18. A forestry department managing a state forest possesses a map of four distinct wildlife management areas and a continuous grid of estimated soil moisture percentages. They need to generate a summary table listing the average soil moisture within each individual area's borders. Deduce the specific class of cell-based spatial operation required for this regional aggregation. Identify the operational role played by each grid layer and determine the exact output value if one area consists entirely of pixels reading zero percent.
19. Based on the types of spatial data queries discussed in class, identify the appropriate type of query for each of the following:
- A city planner wants to identify bus stops located within 500 m of a railway station that serve more than 100 passengers per day.
 - A planning authority wants to identify all property parcels that will be affected by widening an existing road by 10 m on both sides.
 - A GIS analyst wants to determine whether a proposed pipeline crosses any existing railway line.

Section III (10 Marks)

20. A state environmental agency wants to identify priority areas for wetland conservation in a district. You are provided with wetland boundaries, land-use/land-cover data, village boundaries, road networks, and locations of existing protected areas. The agency wants to identify wetlands that are under human-use pressure, located close to settlements or roads, and not currently covered by protected areas. Develop a GIS workflow to derive the priority areas. Explain the sequence of spatial operations you would perform, why each operation is required, and how the intermediate outputs would be used to obtain the final conservation-priority layer.
21. A forest manager wants to assess habitat conditions using raster datasets of forest cover, elevation, and distance from roads. The analysis should first derive a new raster by combining information from the available datasets. It should then examine the characteristics of forest areas in the immediate surroundings of each location. Next, the manager wants to compare forest conditions among different elevation classes. Finally, the manager wants to assess the overall connectivity of forest habitat across the study region.

Mid-Exam

23rd September 2026

Develop a GIS workflow for this analysis. For each stage, identify the appropriate raster function category and justify your choice. Also state one suitable GIS operation for each stage.

(OR)

A city authority has raster datasets of land-surface temperature, vegetation, impervious surface, and population density. The authority wants to identify areas where these variables jointly indicate environmental stress, examine whether individual locations differ from their surroundings, summarise environmental conditions for each municipal ward, and assess the spatial pattern of environmental stress across the city. Design a raster-analysis workflow to address these requirements. Identify the appropriate raster function category at each stage, explain why it is suitable, and mention one suitable GIS operation that could be used.